SOFTWARE REQUIREMENTS SPECIFICATION

HaramBlurr

An Android app that hides bad language and images of people on the screen, in every app.

Version	Date	Scope
2.1 (draft)	6 October 2026	Android, v1 to v4

Status

For team review

1. Introduction 2. Overview 3. Features 4. Quality targets 5. Not planned 6. Status & plan 7. Risks 8. Terms

1. Introduction

1.1 Purpose

This document explains what HaramBlurr must do and how well it must do it. It is written for the project team and supervisors, so it avoids deep technical detail. The full technical study and the test results live in `investigation-android-v1.md` and `p0-results-android-v1.md`.

1.2 The problem

Some people want to avoid bad language and images of people while they use their phone: in comments, chats, social apps, videos, ads and news. Today's tools have three problems:

- Most work only in **one place**, such as one browser or one website.
- Tools that watch the whole screen show the content **first** and cover it **after**, so it flashes for a moment.
- Some send screen content to a server, which is bad for privacy.

1.3 Our idea

HaramBlurr is one Android app. It uses Android's accessibility feature (the same one screen readers use) to know what is on the screen, and it draws covers on top. Everything runs on the phone.

- **Bad language:** the app reads the text that apps show, finds bad words with a word list, and covers them. From v1.1, our own small AI model on the phone adds context: it catches insults without bad words and avoids covering normal uses of words such as "dog".
- **People:** the app looks at a picture of the screen and finds people with an AI model on the phone, then covers them.

1.4 Release plan

VERSION	HIDES	WHY IN THIS ORDER
v1	**Bad-language text** in standard Arabic and English: swearing, sexual words, insults, blasphemy	The easiest start. It builds the app that later versions reuse.
v1.1	**Smarter text filter**: our own small AI model adds context to the word list	Fixes what a word list cannot see; the team's main research contribution
v2	**Big people** (taller than about 96 screen pixels)	The AI model already finds most of them
v3	**Medium people** (about 32–96 pixels)	Needs a closer check in parts of the screen
v4	**Small people** (under about 32 pixels)	Needs our own training on phone screenshots

Each version adds to the one before.

2. Overview

2.1 How it works, simply

Bad language (v1):

- **Listen:** Android tells the app when the screen changes.
- **Read:** the app reads the text on the screen.

- **Clean:** it removes spelling tricks (for example tashkeel, stretched letters, symbols in place of letters).
- **Match:** it looks each word up in the bad-word list.
- **Cover:** it puts a solid block over the bad word or over the whole message (the user chooses).

People (v2–v4): the changed part of the screen is covered at once. The app takes a picture of the other app's window (the picture does not include our covers), the AI model finds the people, and the cover is removed everywhere except over the people.

Content changes	Covered while checked	Shown
scroll, new page, new app	people: about 0.3–0.7 s today	with every detected person hidden

Text needs no "covered while checked" step. The check takes less than a millisecond, so the cover appears right after the text.

2.2 Users

One kind of user: a person who chooses to install the filter on their own phone. It is a personal tool, not a parental-control or monitoring tool.

2.3 Where it runs

ITEM	REQUIREMENT
System	v1 (text): Android 8.0 or newer. v2–v4 (people): Android 14 or newer. Android 13 only with a weaker fallback (FR-16).
Test phone	Xiaomi Redmi 12: MediaTek Helio G88, 2 fast + 6 slow processor cores, small Mali-G52 graphics. A normal low-cost phone.
Internet	Not needed. The app does not even ask for internet access.
Install	From an APK file (not Google Play) for the graduation project

2.4 Assumptions

- The user turns on HaramBlurr in the phone's accessibility settings and allows it to run in the background.
- "Bad language" means the words and phrases on our list, in four categories: swearing, sexual words, insults and blasphemy.

- "Person" means any human figure: real, drawn, animated or in a game.
- Hiding something by mistake is acceptable. Showing a person or a bad word by mistake is not. When in doubt, hide.

3. Features (functional requirements)

Priority: **Must** = needed for that version, **Should** = planned if time allows, **Could** = later.

3.1 The app (built in v1, used by all versions)

ID	REQUIREMENT	VER.	PRIORITY	STATUS
FR-1	Before turning on, show a clear screen that explains what the app can see and that nothing leaves the phone.	v1	Must	Planned
FR-2	Show a small notification while the filter is on. If the filter stops or crashes, the notification goes away and the user can see that.	v1	Must	Planned
FR-3	Covers move with the page when the user scrolls, so covered content stays covered and the cover does not flicker.	v1	Must	Planned
FR-4	Let the user turn the filter on and off and skip chosen apps.	v1	Should	Planned

3.2 Bad-language text (v1)

ID	REQUIREMENT	VER.	PRIORITY	STATUS
FR-5	Read the text shown by every app that shares its text with Android: social apps, chats, browsers, news.	v1	Must	Planned
FR-6	Find bad words and phrases in standard Arabic and English, in four categories: swearing, sexual words, insults, blasphemy.	v1	Must	List needed The Arabic list must be built by the team.
FR-7	Catch common spelling tricks: tashkeel, stretched letters (ـ), different alef and ya	v1	Must	Planned

	forms, repeated letters, symbols in place of letters.			
FR-8	Match whole words only, so a normal word that contains a bad word is not covered.	v1	Must	Planned
FR-9	Cover with a solid block. The user chooses: cover only the bad word, or the whole message.	v1	Must	Planned
FR-10	Let the user turn each category on and off.	v1	Should	Planned
FR-11	Do not cover text the user is typing.	v1	Should	Planned
FR-12	Also read text inside images and videos (memes, thumbnails, subtitles). Planned with PaddleOCR (Arabic mobile model, about 8 MB). Google's on-device OCR does not support Arabic.	later	Could	Later
FR-20	Our own small on-phone AI text model as a second layer: it reads the whole message, decides the category with a confidence, and points at the offensive **words**. It handles context-dependent words and insults without bad words. Trained by the team from large open "teacher" models (investigation §3.5). Compared with the word list and ready-made small models.	v1.1	Should	Planned
FR-21	The user chooses which language(s) to filter: **one by default** (the phone's language), two if needed. Each language is a "pack" with its own word list and, from v1.1, its own model. Only the chosen packs use memory. Text is sent to the right pack by its letters (Arabic or Latin).	v1	Must	Planned

3.3 People (v2–v4)

ID	REQUIREMENT	VER.	PRIORITY	STATUS

FR-13	Detect people in every app that shows images. Cover each one with a strong pixelation or solid fill, a little larger than the person. A light blur still shows the body shape.	v2	Must	Tested offline
FR-14	Cover changed content as soon as the phone reports the change, and keep it covered until it has been checked. The cover never lifts by itself.	v2	Must	Planned
FR-15	Some apps block screenshots (banking, some video apps). The user chooses to leave those apps alone or always cover them.	v2	Should	Planned
FR-16	Fallback for Android 13: use screen capture instead of window pictures. In this mode, new content can show for a moment before it is covered.	v2	Could	Planned
FR-17	Do a closer check in parts of the screen (tiles) to find medium-size people.	v3	Must	Needs work Works offline. Too slow on the phone today.
FR-18	Find small and far-away people, using a model trained on real phone screenshots.	v4	Must	Needs work
FR-19	A special mode for full-screen video that checks more often.	later	Could	Later

4. Quality targets (non-functional requirements)

ID	AREA	TARGET	TODAY
NFR-1	Text safety (v1)	Hide at least 95% of the bad words on our list in real comments and chats.	Not measured
NFR-2	Text false covers (v1)	No more than 1 wrong cover per 1 000 clean words (news, Quran, normal chats).	Not measured

NFR-3	Gap before the cover	New content shows for no more than about 3 frames (about 33 ms on a 90 Hz screen) before it is covered.	**Not measured** First measurement in v1.
NFR-4	People safety (v2–v4)	Hide at least 95% of people of that version's size: big in v2, medium in v3, small in v4.	**Not yet** On test screens, one pass (with tiles): big 85–89% (91–95%), medium 60–76% (87–89%), small 24–27% (51–54%).
NFR-5	Waiting time (people)	Changed content stays covered for no more than about 0.7 s.	**Not measured** Expected 0.3–0.7 s from the parts.
NFR-6	Check speed (people)	One full-screen check takes 150 ms or less on the test phone.	**Not yet** 295 ms on the processor. The faster options are not measured yet.
NFR-7	Smoothness	Touches pass through the covers. The phone does not feel slower.	**By design**
NFR-8	Battery	No work while the screen is still.	**By design** Battery use not measured yet.
NFR-9	Memory	v1 under 50 MB. With the text model (v1.1) under 150 MB. With the people model too, under 250 MB.	**Likely** People model alone: about 98 MB.
NFR-10	Privacy	Screen text and images stay in memory only. Nothing is saved and nothing is sent. The app has no internet permission.	**By design**
NFR-11	Stability	A crash in HaramBlurr must never close or freeze other apps.	**By design**
NFR-12	Permissions	The user grants access **once**. The app never asks again on every use or after a restart. A common complaint about a competing app.	**By design** v1 needs no screen capture. v2+ uses the accessibility window picture, which needs no per-session prompt.

NFR-13	Text model (v1.1)	Per language pack, on the test phone: real RAM at most 100 MB, at most 50 ms per message (p95), about one CPU core at most while checking (or the GPU). Better than the word list alone and than ready-made small models on the same test set.	Not built Candidates are measured on the phone first (step P0c).

5. Not planned

- iPhone. Apple does not let apps see or cover other apps.
- Arabic dialects and Arabizi (Arabic written in Latin letters). Possible after v1.
- A root mode that works inside Android's screen drawing
- The Linux desktop version. It is paused as plan B, and its test results still help.
- Hiding only some people, for example by gender. This is much less accurate, and a mistake means a person is shown.
- Blocking the user from turning the filter off. It is a personal choice, not a lock.

6. Status and plan

STEP	WHAT	TIME	STATUS
v1-a	App skeleton: accessibility service, covers, settings, notification, consent screen	~1 week	Next
v1-b	Text filter: clean, match, locate the word, cover, follow scrolling. Measure the gap.	1–2 weeks	Planned
v1-c	Word lists: build the Arabic list, review the English list, test on real comments	~1 week	Planned
v1.1	Own text model: data, labels from large open models, training, comparison	4–5 weeks	Planned
P0	Test the people model for speed and accuracy on a laptop and on test phone screens	days	Done
P0b	Finish the speed test on the Redmi 12	1 evening	Half done

v2	People model in the app, big people	~2 weeks	Planned
v3	Tiles and a faster model setup, medium people	1–2 weeks	Planned
v4	Train on real phone screenshots, small people	weeks	Planned

What the tests showed so far: the people model finds big people well. Small people and phone speed are the open problems, and they come later (v3, v4). v1 does not depend on them.

7. Risks

RISK	EFFECT	WHAT WE DO
No complete free Arabic word list (the best free one has 348 Arabic-script entries)	Many Arabic bad words are missed	Start from free lists (LDNOOBW V2, Attia's lexicon). The team completes the list, then tests it on real Arabic comments.
Insults and blasphemy depend on context	Normal text gets covered	Phrases instead of single words, a strict insult list, a switch per category
Some apps do not share their text	Bad words in those apps are not covered	List the apps that work. Text from pixels (OCR) later.
Small people are missed	Some people are shown	Tiles, training on phone screenshots, hide when unsure
The phone is too slow	Long waiting time, no tiles	Faster model types, graphics chip, smaller input. Otherwise a stronger test phone.
Xiaomi's battery saver stops the app	The filter stops without warning	A setup screen that asks for Autostart and no battery limits. The notification (FR-2) shows when it stops.

"HaramBlur" already exists as a browser extension and an Android app (500K+ downloads)	Our project must offer something it does not. Name confusion.	Bad-language text, cover only the person, hide first without a timeout, every app. Cite it as related work. Consider a new name.

8. Terms

Frame	One picture on the screen. The Redmi 12 shows up to 90 per second, so one frame is about 11 ms.
ms / s	Millisecond (one thousandth of a second) / second
Accessibility service	A special kind of Android app that the user turns on in settings. It can read what is on the screen and draw on top of it. Screen readers use the same feature.
Cover	The solid block or strong pixelation that we draw over a bad word or a person
Word list	Our list of bad words and phrases, each tagged with a category
Cleaning (normalization)	Removing spelling tricks before matching, so "كلبببب" and "كلب" count as the same word
OCR	Reading text from a picture. Needed for text inside images, which comes later (PaddleOCR).
AI model	The trained program that finds people in a picture
Tiles	Checking the screen in a few parts instead of all at once, so small people look bigger to the model
Hide first	Our rule for people: new content stays covered until it is checked (in technical documents: "fail-closed")